



RETURNSHIELD AI

Return fraud decisioning for modern commerce

01 / 06

PURPOSE-BUILT FOR POST-PURCHASE RISK

Stop return fraud before it becomes refund loss.

Score every return request, explain the decision, and block risky refunds before cash leaves the system.

Built for the fraud surface checkout tools miss: empty-box claims, wardrobing, item swaps, fraud rings, and suspicious return narratives.

ReturnShield AI · product presentation

0-100

RISK
SCORE

3

DECISION
BANDS

Explain

REASON
TRACE

Live

API
SCORING

WHY IT MATTERS

Return fraud happens after checkout, when payment fraud tools stop watching. ReturnShield AI makes the refund moment the decision point.



THE PROBLEM

Legacy fraud stacks miss the returns layer.

THE GAP

Checkout fraud systems are optimized for cards and chargebacks. Returns include shipping, text, images, and account behavior those systems do not fully inspect.

WHAT ATTACKERS EXPLOIT

Empty-box claims, weight fraud, wardrobing, price arbitrage, fake narratives, and organized account rings.

DIMENSION	PAYMENT FRAUD	RETURN FRAUD
Detection point	Checkout	Post-fulfillment
Signals	~50 features	200+ features across order, text, graph, images
Attack surface	Card details	Physical goods, labels, weight, chat, photos
Existing coverage	Strong	Near zero

Problem framing · why the product exists



PRODUCT FEATURES

One workflow
from scoring to
analyst action.

RETURN SCORING

Scores every return before the refund is released.

EXPLAINABILITY

Shows why a request was flagged in plain language.

CASE REVIEW

Routes medium and high-risk returns to analysts.

FEEDBACK LOOP

Stores analyst decisions for retraining and follow-up.

FRAUD RING VISIBILITY

Surfaces connected accounts and repeat behavior.

REASON CODES

Turns score output into actions the team can trust.

DECISION BANDS

0-39

AUTO_APPROVE

Low risk, release the refund.

40-69

MANUAL_REVIEW

Needs analyst inspection.

70-100

HOLD_REFUND

Strong fraud indicators present.

OPERATIONAL EFFECT

Fewer wrongful refunds. Faster handling for good customers.
Better analyst consistency. Less time spent guessing.

Product story · why teams adopt it



ALGORITHM

Fuse rules, ML, text, and anomalies into one decision.

SIGNAL LAYERS



SCORING FORMULA

$$\begin{aligned} \text{final_score} = & \\ & (\text{rule_score} * 0.30) + \\ & (\text{structured_ml_score} * 0.30) + \\ & (\text{nlp_score} * 0.25) + \\ & (\text{anomaly_score} * 0.15) \end{aligned}$$

The goal is not a single clever signal. The goal is a stable control loop that sees enough of the return event to make a defensible decision.



How the model works



ACCURACY

Accuracy comes from layered validation, not black-box claims.

WHAT ACCURACY MEANS

Precision on obvious fraud, recall on subtle fraud, consistency across repeated cases, and reviewability for analysts and auditors.

BUILT-IN SAFEGUARDS

Ensemble scoring, explainability, feedback capture, and thresholding keep the system measurable and easy to tune.

WHAT THE REPO SUPPORTS TODAY

- Score outputs on a 0-100 scale
- Score breakdown by signal type
- Precision, recall, and F1 tracked in training
- Analyst feedback stored for retraining

SIGNAL PICTURE



The repository documents scoring and validation mechanics, but it does not publish audited benchmark percentages. That keeps the pitch credible.

Accuracy story · measurable and explainable



ADOPT FAST

Score, explain,
review, and
learn in one
control loop.

ReturnShield AI turns returns into a measurable
control point for merchants who need to stop refund
loss without slowing legitimate customers.

MERCHANT VALUE

Fewer wrongful refunds,
faster review, better analyst
consistency, stronger
escalation evidence.

SYSTEM OUTPUT

Risk score, decision band,
reason codes, and readable
explanations.

WHAT HAPPENS NEXT

Plug in the API, score live returns, route the risky ones, and
let feedback improve the model over time.

SUGGESTED PILOT

Start with one merchant, one return channel, and one analyst
queue. Measure refund loss prevented and review time
saved.